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University under section 2(f) of UGC Act, 1956

MAJOR PROJECT SYNOPSIS

A Project

Submitted in Partial Fulfilment of the
Requirements for Award of the Degree of
Bachelor of Technology, 4th year

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February 2026



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Project Title Explanation-

This project aims to design and implement an intelligent, cloud-based Ed-Tech platform that provides students and instructors with a centralized, interactive, and scalable learning environment. CleverPath AI focuses on enhancing online education by integrating modern full-stack development with AI-driven assistance and smart content discovery.

o Cloud-based Ed-Tech Platform →

The system is designed to be hosted on the cloud, allowing users to access the platform anytime and from anywhere. Cloud deployment ensures scalability, reliability, and availability without requiring users to manage local servers or infrastructure.

o Ed-Tech Platform →

CleverPath AI functions as an online education system where instructors can create and manage courses, and students can enroll, learn, and interact with educational content. The platform supports structured learning, progress tracking, and feedback, making it suitable for academic and skill-based education.

o AI-powered Assistance →

The platform integrates AI features such as a chatbot for instant doubt resolution and user guidance, along with a smart search mechanism that understands user intent. These features reduce dependency on instructors for basic queries and improve learning efficiency and engagement.

o MERN Stack →

The project is developed using the MERN stack, which ensures a robust and scalable full-stack architecture:

MongoDB → Database used to store users, courses, enrollments, and learning data.

Express.js → Backend framework for handling APIs, authentication, and business logic.

React.js → Frontend library used to build an interactive and responsive user interface.

Node.js → Server-side runtime environment that powers the backend services.



Cloud & Security Infrastructure →

The platform is designed to be cloud-ready with secure authentication and role-based access control. Cloud services can be used for hosting, database management, and secure data storage, ensuring reliability, performance, and data protection.

Introduction

Background

In recent years, online education platforms have become a vital component of the modern learning ecosystem. With the rapid growth of the internet and digital technologies, students increasingly rely on online platforms to acquire new skills, prepare for competitive examinations, placements, and upskill themselves in emerging domains such as web development, data science, and artificial intelligence. These platforms offer flexibility, accessibility, and a wide range of learning resources, making education more inclusive and convenient.

Despite their popularity, many existing online education platforms primarily focus on content delivery rather than intelligent learning support. Most systems provide the same learning material to all users, without considering individual learning needs, skill levels, or goals. As a result, students often struggle to identify relevant courses, follow an effective learning path, and receive timely assistance when doubts arise. The absence of personalized guidance reduces learning efficiency and user engagement.

Students also face challenges in navigating large volumes of educational content due to inefficient search mechanisms. Traditional keyword-based search systems require exact queries, making it difficult for learners to discover appropriate courses or topics when they are unsure of precise terminology. At the same time, instructors require scalable and user-friendly tools to manage courses, upload content, track student engagement, and handle queries efficiently, which many platforms fail to provide adequately.

To overcome these challenges, **CleverPath AI** is proposed as a modern and intelligent Ed-Tech platform that integrates the robustness of the **MERN stack** with **AI-powered chatbot support and smart search functionality**. The platform is designed to enhance the overall learning experience by providing instant doubt resolution, intelligent content discovery, and efficient course management. By combining full-stack web development with practical AI integration, CleverPath AI aims to create a more interactive, responsive, and user-centric learning environment.



Problem Domain

Despite the availability of multiple online education platforms, several limitations continue to affect the overall learning experience. One of the major challenges faced by students is the lack of instant doubt resolution. In most platforms, learners have to wait for instructor responses or rely on external resources, which interrupts the learning flow and reduces productivity.

1. **Lack of Instant Doubt Resolution**-Most existing online education platforms do not provide immediate support for student queries. Learners often have to wait for instructor responses or search through external resources to resolve doubts. This delay disrupts the learning flow and reduces overall efficiency and engagement.
2. **Inefficient Content Discovery Mechanisms**-Traditional education platforms rely heavily on keyword-based search systems, which require exact queries to retrieve relevant content. Students who are unfamiliar with specific terminology struggle to find appropriate courses or learning materials, leading to wasted time and frustration.
3. **Limited Intelligent Assistance and Scalability**-Most platforms lack intelligent assistance to handle repetitive and basic student queries. As a result, instructors become over-dependent resources for minor issues, increasing their workload and limiting the platform's ability to scale efficiently for a large number of users.
4. **Fragmented Learning Experience**-Learning resources such as videos, notes, discussions, and assessments are often spread across multiple tools and platforms. This fragmented approach creates confusion for learners, reduces focus, and negatively impacts the continuity and effectiveness of the learning process.



Proposed Solution

CleverPath AI proposes a comprehensive full-stack Ed-Tech solution designed to address the limitations of existing online learning platforms. The system is developed using the **MERN stack (MongoDB, Express.js, React.js, and Node.js)**, which ensures a robust, scalable, and maintainable architecture. By leveraging modern web technologies, the platform provides a seamless and responsive user experience for both students and instructors.

The platform offers a **centralized learning environment** where students can explore courses, enroll, access learning materials, and track their progress, while instructors can create, manage, and update course content efficiently. This unified structure reduces dependency on multiple external tools and ensures a smooth and organized learning workflow. Secure authentication and role-based access control are implemented to protect user data and restrict access to authorized functionalities. To enhance learner support,

CleverPath AI integrates an **AI-powered chatbot** that provides instant doubt resolution and platform guidance. The chatbot assists students by answering common queries, explaining basic concepts, and helping users navigate the system effectively. This feature minimizes waiting time for instructor responses and improves learning continuity by offering 24×7 assistance.

Additionally, the platform incorporates a **smart search mechanism** that understands user intent rather than relying solely on exact keyword matching. This intelligent search functionality enables learners to discover relevant courses and topics more efficiently, even when they are unsure of precise terms. Combined with a scalable and cloud-ready architecture, the integration of AI features ensures improved accessibility, efficiency, and user engagement across the learning ecosystem.



Objectives

The main objective of the CleverPath AI project is to develop an intelligent and scalable online education platform that enhances the learning experience for both students and instructors. By integrating modern full-stack technologies with AI-driven features, the system aims to overcome the limitations of traditional Ed-Tech platforms. The project focuses on improving accessibility, interaction, and efficiency within a centralized learning environment.

Primary Objectives

1. To design and develop a robust Ed-Tech platform using the MERN stack.
2. To implement secure user authentication and role-based authorization for students and instructors.
3. To enable instructors to create, manage, and update courses efficiently.
4. To allow students to enroll in courses, access learning content, and provide course ratings and feedback.
5. To integrate an AI-powered chatbot for real-time doubt resolution and user assistance.
6. To develop a smart search mechanism that enables intelligent and intent-based content discovery.
7. To ensure system scalability, reliability, and usability for a large number of users.

Literature Review

- Recent studies on online education platforms highlight the growing importance of **personalized learning, instant feedback, and intelligent assistance** in improving student engagement and learning outcomes. Research in **AI-driven education systems** indicates that conversational agents such as chatbots play a crucial role in providing real-time support, answering repetitive queries, and reducing instructor workload. Furthermore, advancements in **Natural Language Processing (NLP)** have enabled the development of intelligent search mechanisms that move beyond traditional keyword-based approaches, allowing systems to understand user intent and deliver more relevant results. These research findings strongly support the integration of AI chatbot and smart search functionalities in modern Ed-Tech platforms like CleverPath AI. Cloud-Based Collaboration:
 1. **Importance of Personalized and Instant Learning Support-** Research emphasizes that students perform better when learning platforms provide instant feedback and personalized guidance rather than static content delivery.



2. **Role of AI Chatbots in Education-** Studies show that AI-powered chatbots enhance learner engagement by offering 24×7 doubt resolution and reducing dependency on instructors for basic queries.
3. **NLP-Based Smart Search Systems-** Literature highlights that NLP-driven search mechanisms significantly improve content discovery by understanding user intent instead of relying solely on exact keyword matching.

Background Study

1. Introduction to Background Study

Existing Systems

Several popular online education platforms such as **Coursera, Udemy, and Unacademy** have significantly contributed to the growth of digital learning by providing high-quality and diverse educational content. These platforms offer courses across various domains and are widely used by students and professionals for skill development and certification. However, their primary focus remains on content delivery rather than intelligent learning support.

One of the major limitations of these platforms is the lack of **real-time doubt resolution**. Students often depend on discussion forums or delayed instructor responses to resolve their queries, which disrupts the learning flow. Additionally, most platforms rely on **basic keyword-based search mechanisms**, making content discovery inefficient when users are unsure of exact search terms. This often leads to difficulty in finding relevant courses or learning materials.

Furthermore, existing systems offer **minimal intelligent interaction** and largely follow a **one-size-fits-all learning approach**, where the same content is delivered to all learners regardless of their individual needs or learning pace. Although these platforms are rich in content and resources, they lack integrated AI-driven assistance and personalized support at the academic project level, which highlights the need for an intelligent platform like CleverPath AI.

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Comparison –

Feature	Traditional Platforms	CleverPath AI
Tech Stack	Varies	MERN Stack
Chatbot Support	Limited / None	AI-powered Chatbot
Search Mechanism	Keyword-based	Smart Intent-Based Search
Scalability	Platform dependent	Highly Scalable
Learning Experience	Content-centric	User-centric

System Methodology

1. Development Methodology

The project follows a **modular and incremental development approach**, consisting of:

1. Requirement Analysis
2. System Design
3. Backend Development
4. Frontend Development



5. AI Feature Integration
6. Testing
7. Documentation

2. System Architecture

The system follows a **Three-Tier Architecture**:

1. Presentation Layer:

- Developed using React.js
- Provides user interface for students and instructors

2. Application Layer:

- Built using Node.js and Express.js
- Handles business logic, authentication, and APIs

3. Data Layer:

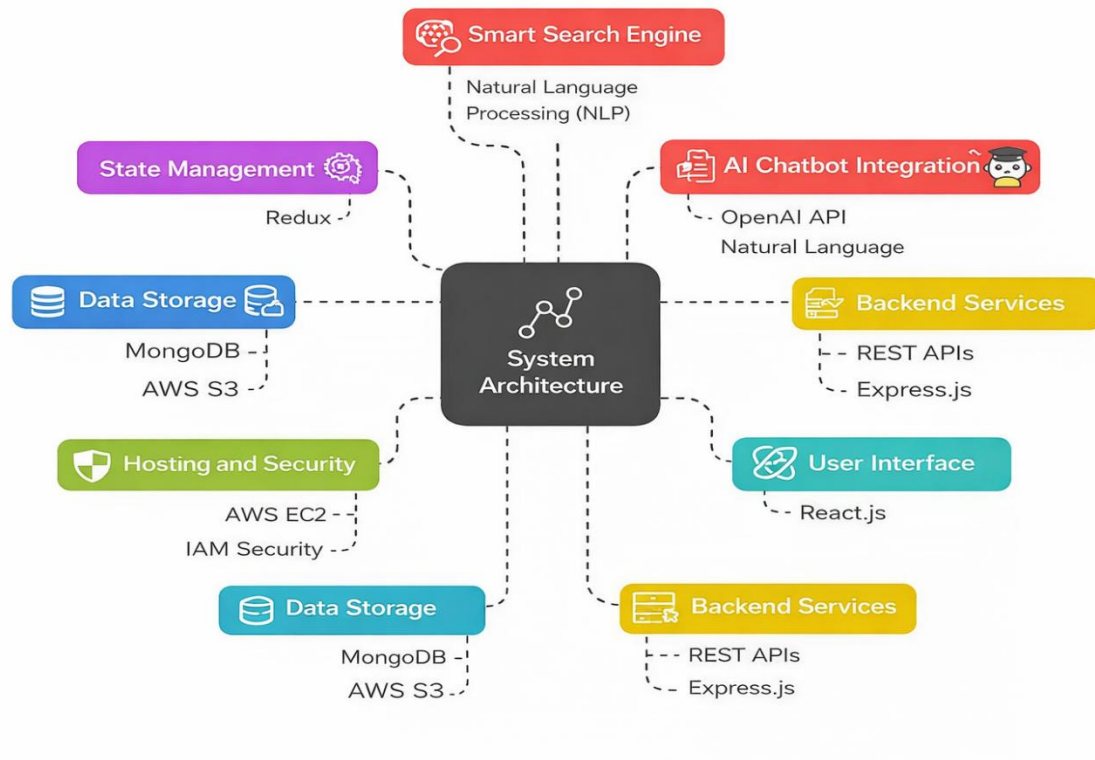
- Implemented using MongoDB
- Stores user data, course data, and interaction history

4. AI Layer

- AI Chatbot for doubt resolution
- Smart Search module using NLP techniques



CleverPath AI System Architecture Overview



3. MODULE DESCRIPTION

3.1 User Management Module

- Registration
- Login
- Authentication



- Authorization
- Role-based access control

3.2 Course Management Module Repository creation

- Course creation and updates
- Enrollment and ratings

3.3 AI Chatbot Module

- Instant doubt resolution
- Platform navigation assistance

3.4 Smart Search Module

- Intent-based search
- Relevant course and content retrieval

3.5 Frontend Module

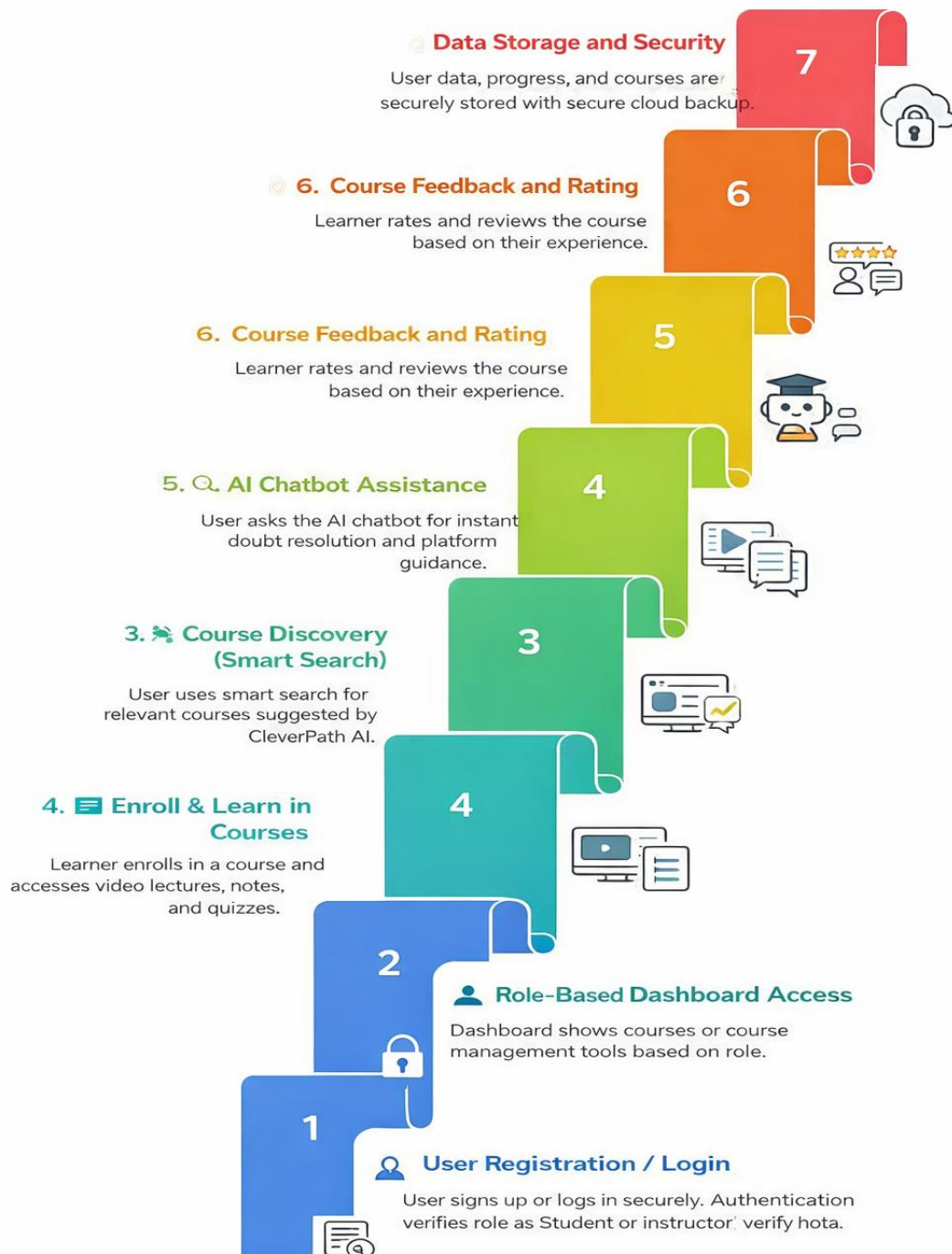
- User dashboard
- Course views.



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CleverPath AI Learning Workflow





RESULTS AND DELIVERABLES

This chapter presents the outcomes obtained after implementing the proposed GitHub-inspired version control system. It highlights the functionality achieved, performance observations, and the final deliverables produced during the development phase of the project.

Results Achieved-

- ❖ The CleverPath AI project has been **partially implemented** as a full-stack Ed-Tech platform using the MERN stack. The frontend and backend modules have been successfully developed and integrated, ensuring smooth communication between the client-side and server-side components. The system currently supports essential functionalities such as user registration, login, course creation, course listing, and content access for students and instructors.
- ❖ The platform demonstrates efficient data handling, responsive design, and a user-friendly interface. Secure authentication and role-based access control mechanisms have been implemented to ensure data privacy and controlled access for different users. These completed modules establish a strong foundation for further enhancement of the system.
- ❖ The AI-powered chatbot and smart search functionalities are planned features of the project and are currently under development. Similarly, the payment gateway integration for paid courses is proposed and will be implemented in the next phase. Once completed, these features will further enhance automation, intelligence, and usability of the platform.'

Deliverables

- ❖ The primary deliverable of the CleverPath AI project is a **partially functional web-based Ed-Tech application** developed using modern web technologies. The current version of the application includes a fully working frontend and backend with core learning and course management features.
- ❖ The complete **source code for the frontend and backend modules** has been developed and organized following modular and scalable design principles. A structured **MongoDB database schema** has been designed to manage users, courses, enrollments, and related data efficiently.
- ❖ In addition, **project documentation** has been prepared, covering system architecture, module descriptions, database design, and implementation details. This documentation provides a clear roadmap for future enhancements such as AI chatbot integration, smart search functionality, and payment gateway implementation.



PROJECT STATUS AND COMPLETION

<u>Phase</u>	<u>Completion</u>
Requirement Analysis	100%
System Design	100%
Backend Development	90%
Frontend Development	85%
AI Integration	10%
Testing(Front+Backend)	60%
Documentation	65%

Overall Completion Status:- Approximately **70% Completed**



LIMITATIONS

- ☐ Chatbot responses depend on available content
- ☐ Limited personalization in the initial phase
- ☐ Advanced analytics not implemented

FUTURE SCOPE

- Advanced AI-based personalization
- ML-driven course recommendations
- Voice-enabled chatbot
- Mobile application support
- Advanced analytics and dashboards